MECHANICAL PROPERTIES OF SOLIDS

PHYSICS FORMULA & NUMERICAL METHODS SHEET • CLASS 11 / JEE / NEET

EXAM REVISION

1. DEFORMING FORCE, STRESS & STRAIN

MUST KNOW

Tensile / Compressive Stress (σ)

BASIC

$$\sigma = \frac{F_{normal}}{A}$$

F: Applied Normal Force (N)

A: Cross-sectional Area (m^2)

SI Unit: N m^{-2} or Pascal (Pa)

Dim: $[M^1 L^{-1} T^{-2}]$

Longitudinal Strain (ε)

BASIC

$$\varepsilon = \frac{\Delta L}{L}$$

ΔL: Change in length (m)

L: Original length (m)

Unit: Dimensionless

Type: Tensile or Compressive

Shearing Stress & Shearing Strain

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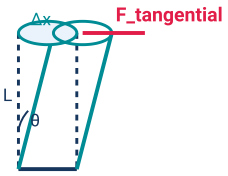
$$\tau = \frac{F_{tangential}}{A}, \quad \theta = \tan(\theta) \approx \frac{\Delta x}{L}$$

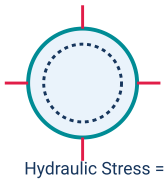
Δx: Lateral displacement

L: Height of cylinder/cube

θ: Shear angle in radians

Condition: Small angle (θ ≤ 10°)





Hydraulic Stress & Volume Strain (ε_V)

MUST KNOW

$$\text{Hydraulic Stress} = p = \frac{F_{normal}}{A}, \quad \varepsilon_V = \frac{\Delta V}{V}$$

p: Applied fluid pressure (Pa)

ΔV: Decrease in total volume

V: Original initial volume

Note: Pure volume change without shape change

CAUTION — COMMON EXAM ERROR: Stress is *NOT* a vector quantity! Although force is a vector, stress at a point in a body has no specific direction in space; it depends on the orientation of the reference area element.

2. HOOKE'S LAW & STRESS-STRAIN CURVE

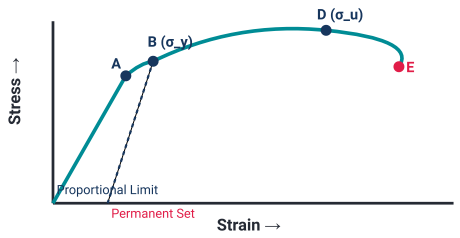
EXAM FAVOURITE

Hooke's Law (Valid strictly within Elastic Limit / Proportional Limit):

For small deformations, internal restoring stress is directly proportional to strain produced:

Stress ∝ Strain ⇒ Stress = k × Strain ⇒ **k = Modulus of Elasticity**

2. STRESS-STRAIN RELATIONSHIP FOR METALS (CONTD.)



Key Regions of Stress-Strain Curve:

- **OA (Linear):** Obeys Hooke's Law. Slope = Young's Modulus (Y).
- **B (Yield Point / Elastic Limit):** Material returns to original size if load is removed below B. Stress = Yield Strength (σ_y).
- **B to D (Plastic Region):** Strain increases rapidly. Material gets permanently deformed (Permanent Set).
- **D (Ultimate Tensile Strength σ_u):** Max load bearing capacity.
- **E (Fracture Point):** Wire breaks. If D & E are close → *Brittle*; if far apart → *Ductile*.

3. MODULI OF ELASTICITY

MUST KNOW

Modulus Type	Formula	Physical Meaning	Applies To
Young's Modulus (Y)	$Y = \frac{F \cdot L}{A \cdot \Delta L}$	Resistance to longitudinal stretching / compression	Solids only
Shear Modulus (G)	$G = \frac{F \cdot L}{A \cdot \Delta x} = \frac{F}{A \cdot \theta}$	Resistance to change in shape (Modulus of Rigidity)	Solids only
Bulk Modulus (B)	$B = -\frac{p}{\Delta V / V}$	Resistance to hydraulic volume compression	Solids, Liquids & Gases
Compressibility (k)	$k = \frac{1}{B} = -\frac{1}{\Delta p} \cdot \frac{\Delta V}{V}$	Fractional volume change per unit pressure increase	Solids, Liquids & Gases

Poisson's Ratio (v)

EXAM FAVOURITE

$$v = -\frac{\text{Lateral Strain}}{\text{Longitudinal Strain}} = -\frac{\Delta d / d}{\Delta L / L}$$

Δd: Change in diameter

d: Original diameter

Theoretical Limit: $-1 \le v \le 0.5$

Practical Range: 0 to 0.5 (Steel ≈ 0.28)

Elastic Potential Energy (U)

MUST KNOW

$$U = \frac{1}{2} \times F \times \Delta L = \frac{1}{2} \cdot \frac{Y A (\Delta L)^2}{L}$$

Energy Density (u): $u = \frac{U}{Volume}$

u Formula: $u = \frac{1}{2} \times \text{Stress} \times \text{Strain}$

In terms of Y: $u = \frac{1}{2} Y (\varepsilon)^2$

SI Unit: Joule (J) / J m^{-3}

SHORTCUT & COMPARISON FACT: Steel is *more elastic* than copper, brass, or aluminum because for an identical force and area, steel undergoes much smaller elongation (ΔL), meaning Young's Modulus $Y_{steel} > Y_{copper} > Y_{aluminum}$. Approximate relation: Shear Modulus $G \approx Y / 3$.

4. KEY DERIVATIONS & STEP-BY-STEP PROOFS

DERIVATION

Derivation of Elastic Strain Energy Stored in Stretched Wire:

Goal: Prove that Potential Energy per unit volume u = $\frac{1}{2}$ × Stress × Strain.

1. Starting Equation: Restoring force for elongation *l* is given by Young's modulus: $F = \frac{Y \cdot A \cdot l}{L}$

2. Work done for small extension dl: $dW = F \cdot dl = \frac{Y A l}{L}\, dl$

3. Total Work Done to stretch from 0 to ΔL:

$$W = \int_0^{\Delta L} \frac{Y A l}{L}\, dl = \frac{Y A}{L} \cdot \left[\frac{l^2}{2} \right]_0^{\Delta L} = \frac{1}{2} \cdot \frac{Y A (\Delta L)^2}{L}$$

4. Final Energy Density: Divide W by Volume (A · L):

$$\mathbf{u = \frac{W}{A \cdot L} = \frac{1}{2} \times \left[\frac{Y \Delta L}{L} \right] \times \left[\frac{\Delta L}{L} \right] = \frac{1}{2} \times Stress \times Strain}$$

5. PRACTICAL APPLICATIONS OF ELASTICITY

EXAM SHORTCUTS

Bending of Beam (Sagging δ)

JEE MAIN

$$\delta = \frac{W \cdot L^3}{4 \cdot b \cdot d^3 \cdot Y}$$

W: Load applied at center

L: Length of beam span

b: Breadth of beam

d: Depth of beam

I-Section Beam: Prevents buckling & maximizes depth d with reduced weight.

Maximum Height of Mountain

CONCEPT

$$h_{max} = \frac{\sigma_{critical}}{\rho \cdot g} \approx 10 \text{ km}$$

$\sigma_{critical}$: Elastic limit of rock (≈ 3×10^8 N m^{-2})

ρ: Density of rock (≈ 3×10^3 kg m^{-3})

g: Acceleration due to gravity (9.8 m s^{-2})

6. NUMERICAL-SOLVING WORKFLOWS

EXAM METHOD

METHOD 1: Elongation of Composite Wires (Equal Tension / Load)

Scenario: Copper (L_c, Y_c) and Steel (L_s, Y_s) wires connected end-to-end pulled by load W.

Given: L_c, L_s, Y_c, Y_s, Total ΔL_{net} → **Ratio:** $\frac{\Delta L_c}{\Delta L_s} = \frac{Y_s}{Y_c} \cdot \frac{L_c}{L_s}$ → **Solve:** $\Delta L_c + \Delta L_s = \Delta L_{net}$ → **Find W:** $W = \frac{A \cdot Y_c \cdot \Delta L_c}{L_c}$

METHOD 2: Minimum Radius of Crane Wire Rope

Scenario: Lift mass M without exceeding yield strength (σ_y) with safety factor N.

Required Force: F = N × M × g → **Area:** $A_{min} = \pi r^2 = \frac{F}{\sigma_y}$ → **Radius:** $r = \left(\frac{N \cdot M \cdot g}{\pi \cdot \sigma_y} \right)^{1/2}$

7. PHYSICAL CONSTANTS & MATERIAL PROPERTIES TABLE

NCERT DATA

Material	Density ρ (kg m^{-3})	Young's Modulus Y (10^9 N m^{-2})	Yield Strength σ_y (10^6 N m^{-2})	Shear Modulus G (10^9 N m^{-2})
Steel	7860	200	250	84
Copper	8890	110	200	42
Aluminium	2710	70	95	25
Iron (Wrought)	7800	190	170	70
Glass	2190	65	50	23
Water	1000	— (Bulk B = 2.2 GPa)	—	—

8. COMMON EXAM TRAPS & CALCULATION ERRORS

CAUTION

1. Radius vs. Diameter Errors: Area is $A = \pi r^2 = \pi (d/2)^2 = \frac{\pi d^2}{4}$. Forgetting to divide diameter by 2 leads to a 4x calculation error in Stress and Young's Modulus!

2. Unit Conversions:

- 1 GPa = 10^9 Pa = 10^9 N m^{-2}
- 1 MPa = 10^6 Pa = 10^6 N m^{-2}
- 1 mm^2 = 10^{-6} m^2 | 1 cm^2 = 10^{-4} m^2
- 1 atm = 1.013 × 10^5 Pa

3. Sign in Bulk Modulus: Bulk Modulus formula $B = -\frac{\Delta p}{\Delta V / V}$ has a negative sign because an increase in pressure ($\Delta p > 0$) causes a decrease in volume ($\Delta V < 0$), keeping B positive.

9. HIGH-YIELD REVISION FORMULA SUMMARY

QUICK REVISION

Thermal Stress Formula:
$F / A = Y \cdot \alpha \cdot \Delta T$
(α = coeff. of linear expansion, ΔT = temp change)

Elongation under Self-Weight:
$\Delta L_{self} = \frac{\rho \cdot g \cdot L^2}{2Y} = \frac{M \cdot g \cdot L}{2AY}$
(Exactly half of elongation due to end load M)